Before you do anything

Complete these three steps first.

✓ **Docker Desktop is running**	Open it first; mlai-ucb-capstone shows green
✓ **Groq or OpenAI key in .env**	Required by the API; one provider is enough
✓ **uv and Node.js installed**	Backend and frontend tooling

Capstone

Instructions for Jessica Cervi

Startup steps

From clone to running app in three commands

Qdrant

API :8000

UI :5173

Quick start: 3 steps

1

Copy the env template

```
cp .env.example .env
```

2

Add your Groq or OpenAI key

```
GROQ_API_KEY=<your key>   # or OPENAI_API_KEY
```

3

Run everything

```
./start.sh
```

Choose the LLM provider

Groq is the default. Change it in .env, then restart ./start.sh.

1. **Use Groq (default)**

```
LLM_PROVIDER=groq
```

2. **Use OpenAI**

```
LLM_PROVIDER=openai
```

3. **Run more calls at once**

```
LLM_WORKERS=24   LLM_MAX_CONCURRENT=24
```

Also set OPENAI_API_KEY. OpenAI can go up to 24 workers; keep Groq at 4 (it limits output per minute). Whole Constitution: OpenAI about 7 min, Groq about 17 min.

What ./start.sh launches

1

Qdrant

Vector DB in Docker

`localhost:6335`

2

API

FastAPI / uvicorn

`localhost:8000`

3

UI

Frontend dev server

`localhost:5173`

Open the app at http://localhost:5173

Manual start (3 terminals)

1

```
docker compose up -d
```

Qdrant on localhost:6335

2

```
uv run uvicorn capstone.main:app --reload
```

API on :8000. First start downloads the models

3

```
cd frontend && npm install && npm run dev
```

UI on http://localhost:5173

Then run ingestion

The knowledge base is empty (Library: 0) until documents are ingested.

1. **Default document**

```
data/docs/Constitution of India (2024).pdf
```

2. **Open the Documents page**

```
http://localhost:5173  >  Documents
```

3. **Click the button**

```
Ingest the docs folder
```

Reads every file in the server's DOCS_DIR. Upload also accepts PDF, DOCX or TXT. "Not Found" / "API offline" means start the API first.